

Board-Free Multi-LiDAR Extrinsic Calibration for UAV-Anchored Roadside Sensing: Active Attitude Excitation as a Necessary Condition

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Abstract—Board-free extrinsic calibration of roadside LiDARs from a UAV body is governed not by sensor noise but by *flight attitude diversity*. The empirical body centroid is an aspect-dependent bias vector field on the relative bearing sphere u_B ; under concentrated aspect it becomes indistinguishable from extrinsic translation and silently pollutes estimates (~ 211 mm in simulation), while active 3D attitude excitation restores identifiability and decimeter-level accuracy (~ 20 mm in simulation). We present a separable two-stage estimator with per-LiDAR independent Stage-2 solves for zero-FOV-overlap diagonal deployment, an aspect gate that falls back when excitation is insufficient, and a relative-extrinsic covariance analysis showing coherent RTK bias cancels in the relative frame ($0.166\times$ in simulation, scoped). Field validation on a diagonal dual-Ruby intersection is structured but pending.

Index Terms—LiDAR calibration, extrinsic calibration, UAV, RTK, board-free calibration, multi-sensor systems

I. INTRODUCTION

One would expect the accuracy of a board-free calibration to be governed by LiDAR range noise and the centimeter-level fixes of RTK. It is not. Accuracy is instead set by a quantity rarely treated as a design variable: how *diversely the UAV attitude is excited* during the calibration flight—*board-free calibration accuracy is determined by flight attitude diversity, not by the sensor*.

The body centroid used for board-free alignment is not a fixed offset. It is an *aspect-dependent bias vector field* on the relative bearing u_B . When aspect is concentrated, this field aligns with extrinsic translation until the two become *indistinguishable*: a structural degeneracy that silently corrupts estimates (~ 211 mm in simulation) and that joint lever-arm optimization cannot remove. Only active 3D attitude excitation—sector POI tidal-lock combined with pitch and yaw diversity—decorrelates the field so its world-frame mean can be estimated and subtracted, recovering ~ 20 mm accuracy in simulation. Active attitude excitation is therefore a **necessary condition** for unbiased board-free calibration: a flight-design requirement, not a refinement.

Cooperative perception at urban intersections increasingly relies on multiple roadside LiDARs on diagonal posts with ~ 70 m baselines and negligible field-of-view overlap—settings where calibration boards and shared views are impractical. A UAV carrying RTK and a body-mounted LiDAR naturally threads the combined visibility of all sensors, enabling *board-free* calibration from vehicle point clouds alone. In this regime the attitude-excitation finding is not academic: without delib-

erate flight diversity, sub-decimeter extrinsics are structurally unattainable regardless of sensor grade.

A. Contributions

We contribute:

- 1) A **separable two-stage framework**: Stage-1 fits an RTK-anchored body trajectory $T_{WB}(t)$; Stage-2 freezes the trajectory and calibrates each LiDAR extrinsic in an independent solve—enabling diagonal multi-LiDAR deployment with zero FOV overlap.
- 2) A **board-free identifiability mechanism** (core contribution): we show that aspect-dependent centroid bias and extrinsic translation become collinear in the Fisher information under concentrated attitude, yielding a structural non-identifiability that no joint estimator can break; active 3D attitude excitation restores identifiability, and an aspect gate provides a principled fallback when excitation is insufficient (simulation).
- 3) A **relative-extrinsic common-mode dichotomy**: under a coherent systematic RTK bias assumption, $\text{Cov}(T_{\text{rel}})$ scales as $0.166\times$ absolute covariance, whereas white per-epoch noise composes at $3.101\times$ (simulation, scoped).

We structure the field validation protocol (diagonal dual-Ruby intersection) and will report results in Sec. VIII.

Sphere targets (scope boundary). Retroreflective spheres appear *only* as an independent ground-truth ruler to score board-free error in field experiments. They are **not** a fallback calibration method; board-free remains the sole calibration pipeline.

II. RELATED WORK

A. Continuous-time and spline-based calibration

Continuous-time trajectory representations coupled with batch factor-graph optimization are standard for multi-sensor extrinsic calibration [1], [2]. Our Stage-1 trajectory fit follows this lineage but *separates* trajectory estimation from per-sensor Stage-2 extrinsic refinement, avoiding cross-sensor optimization coupling that can leak state between LiDAR solves.

B. Roadside, V2X, and multi-LiDAR systems

Infrastructure sensing at intersections motivates long-baseline, low-overlap multi-LiDAR layouts [TODO: cite: **roadside V2X multi-LiDAR calibration**]. Prior work often assumes overlapping views or shared calibration targets. We address

the diagonal-post regime (~ 70 m baseline, negligible FOV overlap) with UAV-RTK anchoring and per-sensor separable solves.

C. Target-based vs. targetless calibration

Target-based LiDAR–camera and LiDAR–IMU calibration remains the industrial default [TODO: cite: target-based LiDAR extrinsic calibration survey]. Targetless methods exploit scene structure or motion constraints; body-centric board-free calibration uses the vehicle hull as a primitive without a physical board. **Distinction:** we do not propose sphere targets as an alternative pipeline; spheres are reserved for field ground-truth evaluation only (Sec. VIII).

D. UAV-assisted and cooperative calibration

UAV platforms with RTK positioning enable mobile anchoring of roadside sensors [TODO: cite: UAV RTK-assisted sensor calibration]. Related trajectory-matching approaches align sensor frames via shared motion [TODO: cite: trajectory-matching extrinsic calibration]. Our contribution is orthogonal: even with a perfect RTK trajectory, board-free calibration fails without sufficient attitude excitation—a *flight-design* requirement not captured by trajectory matching alone.

E. Relation to prior work on degeneracy

Anonymous prior work on coplanar multi-LiDAR setups identifies *geometric rank deficiency* in joint Fisher information when sensors and the body path are nearly coplanar (simulation $\lambda_{\min} \approx 0$ vs. multi-layer $\lambda_{\min} = 1.074$) [TODO: cite: prior coplanar multi-LiDAR FIM degeneracy]. **This paper addresses a different degeneracy:** aspect-concentrated body-centroid bias aligns with extrinsic translation (~ 211 mm silent pollution in simulation), which joint lever-arm optimization cannot remove (Sec. IV-B). The two failure modes are complementary: coplanar geometry vs. aspect–attitude bias–translation collinearity (Sec. IV-E).

III. SEPARABLE TWO-STAGE CALIBRATION FRAMEWORK

A. Problem setup

Consider L LiDARs rigidly mounted on a UAV body frame $\{B\}$. Each LiDAR ℓ has unknown extrinsic $T_{L\ell B} \in SE(3)$. During a calibration flight, RTK fixes the world pose trajectory $T_{WB}(t)$ and each LiDAR observes body-surface points. We seek $\{T_{L\ell B}\}$ and a consistent $T_{WB}(t)$ without calibration boards.

B. Stage-1: RTK-anchored trajectory fitting

Stage-1 estimates a continuous-time body trajectory $\hat{T}_{WB}(t)$ from RTK position measurements (and optional attitude streams), using Prais–Winsten differenced whitening at high rate (Sec. VI). The output trajectory is **frozen** before any extrinsic refinement.

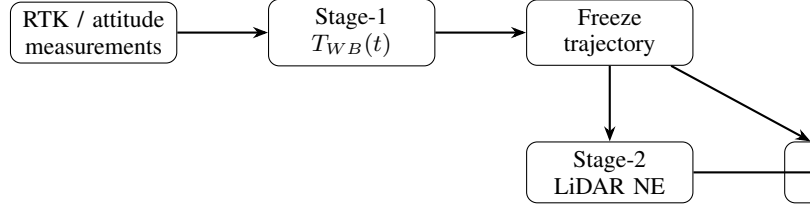


Fig. 1. Separable two-stage pipeline: shared Stage-1 trajectory, independent per-LiDAR Stage-2 solves, post-hoc relative composition. [TODO: replace with publication-quality figure].

C. Stage-2: frozen-trajectory extrinsic refinement

Given frozen $\hat{T}_{WB}(t)$, Stage-2 minimizes a nonlinear least-squares cost over $T_{L\ell B}$ using body-cluster centroid factors and, when the aspect gate opens, an observed-mean lever $\mathbf{p}_B$ (Sec. IV). Each LiDAR is solved in an **independent** optimization scope with no shared state across sensors, preventing cross-sensor leakage in multi-LiDAR batches.

D. Multi-LiDAR composition without FOV overlap

For diagonal deployment (e.g., NE and SW corner posts), sensors need not see the same scene. Per-sensor absolute extrinsics $\hat{T}_{L\ell W}$ are estimated separately; the relative transform is composed post hoc:

$$\hat{T}_{L_{SW} \leftarrow L_{NE}} = \hat{T}_{L_{SW} W} \hat{T}_{L_{NE} W}^{-1}. \quad (1)$$

In simulation, production multi-LiDAR NE/SW errors match single-sensor solves (12.149 mm / 42.041 mm).

IV. BOARD-FREE MECHANISM: ASPECT-DEPENDENT BIAS ON u_B

This section explains why board-free calibration succeeds or fails.

A. Bias as a vector field on relative bearing

At each body-cluster epoch i , define the empirical centroid bias

$$\mathbf{b}_{B,i} = \hat{\mathbf{c}}_{B,i}^{\text{emp}} - \mathbf{L}_{B \rightarrow \text{body}}^{\text{nom}}, \quad (2)$$

where $\hat{\mathbf{c}}^{\text{emp}}$ is the centroid of *visible* sampled points (not the nominal box center plus noise). The relative bearing u_B lives on the unit sphere; $\mathbf{b}_B$ is a **vector field** over flown aspects, not a constant offset. The observed-mean lever $\mathbf{p}_B$ estimates the **mean of this field** over the aspect distribution actually flown:

$$\mathbf{p}_B \approx \mathbb{E}_{u_B} [\mathbf{b}_B(u_B)]. \quad (3)$$

B. Degeneracy as an identifiability statement

Each epoch contributes a world-frame bias driver

$$\mathbf{v}_i = R_{LW} R_{WB}(t_i) \mathbf{b}_{\text{const}}, \quad (4)$$

where $\mathbf{b}_{\text{const}}$ captures the aspect-averaged body-frame bias structure and R_{LW} maps LiDAR to world. Consider the mean over the calibration segment:

$$\bar{\mathbf{v}} = \frac{1}{N} \sum_{i=1}^N R_{LW} R_{WB}(t_i) \mathbf{b}_{\text{const}} \approx R_{LW} \bar{R}_{WB} \mathbf{b}_{\text{const}} \quad \text{when } \{R_{WB}(t_i)\} \quad (5)$$

TABLE I
ASPECT-BIAS COUPLING ON FLIGHT-D (SIMULATION).

Sensor	u_B az std [°]	varying RMS [mm]	$r(u_B, \text{bias}_x)$	$r(u_B, \text{bias}_y)$
NE	14.1	110	+0.53	-0.68
SW	17.6	124	-0.64	+0.69
near-field ref.	—	22	—	—

Concentrated attitude $\Rightarrow$ structural non-identifiability. When attitude is nearly constant ($R_{WB}(t_i) \approx \bar{R}_{WB}$), $\bar{\mathbf{v}}$ is a *fixed* world vector. In the centroid residual, a constant world offset is indistinguishable from a constant shift of the extrinsic translation $\mathbf{t}_{LW}$: both enter the observation model as an identical rigid translation of the projected body cloud. They are therefore collinear in the Fisher information matrix, creating a near-zero eigenvalue—this is not a noise-floor issue but a **structural non-identifiability**. No optimizer, including joint estimation of $\mathbf{p}_B$, can separate bias from translation from data that excite only this degenerate direction; the concentrated-aspect joint-opt path is a no-op at ~ 211 mm in simulation precisely because it cannot break this collinearity.

Diverse attitude $\Rightarrow$ identifiability restored. When $\{R_{WB}(t_i)\}$ spans sufficient pitch and yaw, each epoch rotates $\mathbf{b}_{\text{const}}$ into a different world direction. Summation averages the horizontal components of $R_{WB}\mathbf{b}_{\text{const}}$ (~ 220 mm $\rightarrow \sim 51$ mm mean horizontal norm in simulation)—diversity in *attitude*, not translation alone. A body- Z component ~ 19.5 mm remains because yaw cannot average it; observed-mean $\mathbf{p}_B$ (Eq. 3) explicitly estimates and removes this residual. Attitude diversity is therefore not merely empirically helpful: it is what restores the identifiability of extrinsic translation in the presence of an aspect-dependent centroid bias. **Active 3D attitude excitation is a necessary condition for unbiased board-free calibration.**

C. Engineering consequence: aspect gate and guardrails

When excitation is insufficient, the system must not pretend $\mathbf{p}_B$ is identifiable. An aspect gate ($\text{yaw}_{\text{cv}} \geq 0.65$, with a POI tidal-lock path) attempts observed-mean refinement only when attitude support is adequate; otherwise it falls back to centroid-only. A simulation ground-truth guard further applies observed-mean only if optimization cost and translation norm do not worsen relative to the centroid solution. On Flight-Y (low $\text{yaw}_{\text{cv}} = 0.32$), the gate closes and the estimate reverts to centroid-only at 165.0 mm (simulation)—not a 1243 mm blow-up. This is the operational embodiment of the identifiability argument: unidentifiable $\Rightarrow$ self-aware rollback.

D. Empirical confirmation (simulation)

The identifiability picture above is supported by frozen simulation profiles (Sec. VII).

1) *Aspect-bias coupling (Flight-D)*: Wide u_B sweep with decoupled orbit attitude:

Opposite signs on x vs. y indicate near-sinusoidal coupling to u_B azimuth.

2) 2×2 *aspect-attitude table*:

TABLE II
DEGENERACY TABLE: u_B SWEEP VS. POI-LOCK $\times$ R_{WB} DIVERSITY (SIMULATION).

R_{WB} sweeps		R_{WB} near-locked
NE obs 60.3 / SW 89.9 mm; rel 311 mm	—	—
u_B POI-locked (Flight-W)	NE 12.2 / SW 42.0 mm [†] ; rel rot 0.12°	Flight-J gate OFF: 31.5 mm

[†]SW 42 mm = sector geometric anisotropy (Sec. VII).

TABLE III
CONCENTRATED VS. DIVERSE ASPECT (SIMULATION).

Path	$\ \text{trans}\ $ [mm]	Gate	Note
Centroid-only, concentrated	211.4	—	bias $\parallel \mathbf{t}_{LW}$
Centroid-only, diverse	63.5	—	partial yaw help
Observed-mean $\times 3$, diverse	18.5	ON	$\mathbf{p}_B$ decorrelates
Joint-opt $\mathbf{p}_B$, concentrated	~ 211	—	no-op
Observed-mean, diverse	20.4	ON	production target

3) Concentrated vs. diverse aspect:

4) World-frame averaging under attitude diversity:

Tidal-lock audit (simulation): near-field high-aspect profile yaw-orbit std 0.0° (locked); Flight-D sector-0: 110.4° (not locked).

E. Relation to coplanar geometric degeneracy

Prior work on coplanar multi-LiDAR layouts reports a distinct near-zero Fisher direction: when sensors and the body path are nearly coplanar, joint FIM rank deficiency yields $\lambda_{\min} \approx 0$ versus $\lambda_{\min} \approx 1.074$ under multi-layer flight (simulation; Sec. VII). **That degeneracy is geometric**; the aspect-concentration degeneracy of Sec. IV-B arises from **target bias structure** (bias collinear with translation under fixed attitude). The 2×2 table (Table II) diagnoses the latter; the $\lambda_{\min}$ audit diagnoses the former. Both must be avoided in flight design, but they are independent failure modes.

V. RELATIVE EXTRINSIC UNCERTAINTY

Absolute extrinsics $\hat{T}_{LW}$ inherit RTK position uncertainty. For long-baseline diagonal layouts, the **relative** transform $\hat{T}_{L_{SW} \leftarrow L_{NE}}$ is often the quantity of interest for fusion. We analyze how RTK error modes propagate through independent Stage-2 solves (simulation).

A. Composition and metrics

Relative transform is composed post hoc (Sec. III). We report:

- **center-reg (obs)** — headline metric, rotation-dominated, no 70.7 m lever amplification (40.9 mm, simulation);
- **rel-trans (obs)** — deprecated lever metric (49.0 mm, simulation).

[TODO: Fig. 2 mechanism panel: (a) u_B sphere vector field + Flight-D scatter $r \approx \pm 0.5\text{--}0.7$; (b) 2×2 degeneracy table; (c) $\|\text{mean}(\mathbf{v})\|$ curve 220→51 mm + 19.5 mm vertical bar]

Fig. 2. Board-free mechanism at a glance (placeholder layout). Panel (a): aspect-dependent bias on u_B . Panel (b): degeneracy table. Panel (c): frame-wise world-frame averaging under attitude diversity.

TABLE IV
WORLD-FRAME MEAN BIAS NORM VS. FLIGHT REGIME (SIMULATION).

Regime	$\ \text{mean}(\mathbf{v})\ _h$ [mm]	pitch std [°]	yaw_cv
Concentrated aspect	~ 220	~ 0	~ 0
Diverse near-field	~ 51	~ 16	~ 0.78

TABLE V
RTK INJECTION CAUSAL CHAIN (SIMULATION).

Injection	$\text{tr Cov}(\delta t_{\text{rel}})/\text{mean Cov}(\delta t_{\text{abs}})$	$r(\ \delta p\ _{\text{NE}}, \ \delta p\ _{\text{SW}})$
White noise (per epoch)	3.101 $\times$	0.40
Coherent system bias	0.166 $\times$	1.00

TABLE VI
COHERENT-BIAS REL/ABS RATIO VS. OVERLAP (SIMULATION).

Geometry	coherent rel/mean(abs)
Serial Flight-W	0.166 $\times$
+10 s overlap at handoff	0.15 $\times$

TABLE VII
STAGE-1 RTK PATHS (SIMULATION).

Path	Mechanism	10 Hz antenna RMSE [mm]
v2 uniform AR(1) scaling only	uniform AR(1) scaling only	36.0
PW differenced whitening	PW differenced whitening	27.3
0.5 Hz (PW auto-off, $\Delta t \geq 0.2$ s)	0.5 Hz (PW auto-off, $\Delta t \geq 0.2$ s)	23.7

B. RTK perturbation dichotomy (simulation)

Monte Carlo ($N=20$, Flight-W 0.5 Hz, independent Stage-2 per LiDAR):

Claims (simulation, scoped):

- Under a **coherent systematic-bias assumption**, relative covariance is tighter than absolute (0.166 $\times$).
- White RTK noise **composes** in the relative frame (3.101 $\times$)—not an architecture defect.

C. Temporal overlap at sector handoff (simulation)

Brief temporal overlap can tighten relative UQ without spatial FOV overlap.

D. Scope boundary (mandatory)

The 0.166 $\times$ ratio depends on a **rigid global bias injection** ($\sigma_h=50$ mm, $\sigma_v=100$ mm, one draw applied identically to every epoch in a segment). Real RTK includes drift, colored noise, and lever-arm coupling—**not** perfect common-mode. **Do not cite 0.166 $\times$ as a field guarantee.** Field cancellation is expected to be **weaker**; hardware relative-extrinsic residual under coherent bias is the empirical test (Sec. VIII, Table XII).

VI. PRAIS–WINSTEN RTK WHITENING (STAGE-1)

High-rate RTK (10 Hz) exhibits serial correlation in position residuals. Stage-1 must whiten without destroying low-frequency observability.

A. Why uniform Σ inflation fails

Uniform covariance scaling (wrong AR(1) shortcut) inflates all frequencies and yields ~ 125 mm RMSE—unacceptable (simulation audit, rejected).

B. PW differenced whitening

We apply Prais–Winsten (PW) differenced whitening: estimate lag-1 correlation ρ from RTK residuals and decorrelate successive epochs. Results (simulation, Stage-1 antenna RMSE):

PW reduces 36.0→27.3 mm at 10 Hz; at 0.5 Hz independent epochs PW disables automatically ($\Delta t \geq 0.2$ s). This section supports trajectory quality for Stage-2 but does not replace the attitude-excitation requirement of Sec. IV.

VII. SIMULATION VALIDATION

All numbers in this section are **(simulation)** unless noted. Code, configurations, and frozen simulation baselines will be released upon publication.

We refer to simulation flight profiles as Flight-D (wide aspect sweep with decoupled orbit attitude), Flight-W (POI tidal-lock dual-sector serial), Flight-J (gate-off low-excitation control), and Flight-Y (insufficient yaw diversity guard case).

A. Single-LiDAR near-field high-aspect profile

Diverse aspect ($\text{yaw_cv} \approx 0.78$, $\text{pitch_std} \approx 16^\circ$) enables $\mathbf{p}_B$; centroid floor ~ 64 mm, observed-mean $\sim 18\text{--}20$ mm.

TABLE VIII
SINGLE-LiDAR BOARD-FREE, NEAR-FIELD HIGH-ASPECT PROFILE
(SIMULATION).

Metric	Value [mm]
Centroid-only $\ trans\ $	63.5
Observed-mean $\times 3$	18.5
Joint-opt reference	20.4
GT varying RMS (bias field)	~ 22

TABLE IX
DUAL-LiDAR @ 0.5 Hz (SIMULATION).

	NE obs [mm]	SW obs [mm]	rel rot [°]	center-reg [mm]
Frozen gate	12.2	42.0	0.12	41
Production pipeline	12.149	42.041	—	—

B. Diagonal dual-LiDAR @ Flight-W

Geometry: NE/SW posts at $(\pm 25, \pm 25)$ m, baseline 70.7 m, POI serial lock $t \in [0, 45) \rightarrow \text{NE}$, $t \in [45, 90) \rightarrow \text{SW}$.

1) *Honest NE/SW asymmetry*: SW ~ 42 mm is **confirmed sector geometric anisotropy**, not cross-sensor estimator leakage: independent per-sensor Stage-2 solves reproduce frozen values. Mechanism: SW sightline vs. body main tilt axis $\Rightarrow$ larger lever projection of $\mathbf{b}_{\text{const}}$. **Paper obligation**: present both sectors; do not claim symmetric accuracy.

C. Observability gate (joint FIM $\lambda_{\min}$)

Distinct from aspect bias–translation collinearity (Sec. IV-B): coplanar geometry yields rank deficiency in joint FIM; multi-layer flight restores observability (Sec. IV-E).

D. Two-stage pipeline sanity (sphere-target configuration)

Centimeter-level split 50/50; mean $|LW|$ translation 6.556 mm (simulation).

VIII. FIELD EXPERIMENTS

Status: structured placeholders only — no field numbers in this draft.

This section defines what we will measure on real intersection data. Sphere targets are used **only** as an independent ground-truth ruler to score board-free error; they are **not** a calibration fallback.

A. Experimental setup (placeholder)

- **Site**: urban intersection, diagonal dual Ruby LiDAR posts, baseline ~ 70 m, heights 3–6 m, negligible FOV overlap. [TODO: field campaign A: actual site name, GPS coordinates, post survey]
- **Platform**: DJI M3E + RTK + PSDK LiDAR @ 10Hz. [TODO: field campaign A: firmware versions, antenna lever-arm survey]
- **Flight**: POI tidal-lock sectors per simulation Flight-W; attitude diversity per near-field high-aspect recipe. [TODO: field campaign B: flight logs, POI sector timing, weather]

TABLE X
COPLANAR ABLATION VS. MULTI-LAYER (SIMULATION).

Configuration	$\lambda_{\min}$
Multi-layer (well observed)	1.07357 (≈ 1.074)
Coplanar (degenerate)	≈ 0 (numerical null)

TABLE XI
FIELD BOARD-FREE ACCURACY VS. SPHERE-ANCHOR GROUND TRUTH.

LiDAR	$\Delta trans$ [mm]	Δr
NE	[TODO: field campaign A: NE $\Delta trans$ vs sphere-anchor]	[TODO: field campaign A: NE Δr vs sphere-anchor]
SW	[TODO: field campaign A: SW $\Delta trans$ vs sphere-anchor]	[TODO: field campaign A: SW Δr vs sphere-anchor]

- **Ground truth**: retroreflective sphere anchors surveyed by total station (independent of board-free pipeline). [TODO: field campaign C: sphere positions, survey uncertainty]

B. Sphere-anchor evaluation protocol (method, not calibration)

On the **same** flight segment:

- 1) Survey sphere centers (total station) $\Rightarrow$ ground-truth extrinsic ruler.
- 2) Run board-free calibration once (sole method).
- 3) Report $\Delta trans$, Δrot , center-reg vs. sphere-derived truth.

Spheres measure error; they do not replace board-free.

C. Results (placeholder)

D. Research questions for field (results pending)

- 1) **Real point-cloud bias magnitude**: does empirical $\mathbf{b}_B(u_B)$ on live data match simulation field shape? [TODO: field campaign B: per-epoch bias vs u_B scatter plot data]
- 2) **Board-free field accuracy**: sub-decimeter NE and honest SW sector asymmetry at intersection scale? [TODO: field campaign A: Table XI cells]
- 3) **Coherent-bias residual**: is field rel/abs ratio weaker than $0.166\times$? [TODO: field campaign D: Table XII cells]
- 4) **POI flight feasibility**: can operators execute tidal-lock sectors without degrading RTK or safety margins? [TODO: field campaign B: POI execution notes + gate yaw_cv statistics]

IX. CONCLUSION

We presented a separable RTK-anchored framework for board-free multi-LiDAR extrinsic calibration at intersections with negligible FOV overlap. The central result is an identifiability argument: aspect-dependent body-centroid bias becomes collinear with extrinsic translation under concentrated attitude, producing structural non-identifiability (~ 211 mm in simulation) that no joint optimizer can break; active 3D attitude excitation restores identifiability and ~ 20 mm accuracy (simulation). An aspect gate operationalizes this insight by rolling back when excitation is insufficient. Relative extrinsic

TABLE XII
FIELD TEST OF COHERENT-BIAS CANCELLATION IN RELATIVE EXTRINSICS
(VS. SIMULATION $0.166\times$).

Metric	Simulation	Field
$\text{tr Cov}(\delta t_{\text{rel}})/\text{mean Cov}(\delta t_{\text{abs}})$	$0.166\times$	[TODO: field campaign D: measured rel/abs ratio under real RTK]
$r(\ \delta p\ _{\text{NE}}, \ \delta p\ _{\text{SW}})$	1.00 (coherent)	[TODO: field campaign D: epoch correlation]
center-reg under coherent perturbation	—	[TODO: field campaign D: center-reg residual]

[TODO: field campaign A: intersection layout +
diagonal Ruby posts + flight path]

Fig. 3. Field deployment schematic (placeholder).

uncertainty under coherent RTK bias can be tighter than absolute ($0.166\times$, simulation, scoped assumption); white noise composes at $3.101\times$.

Field validation on diagonal dual-Ruby data is structured in Sec. VIII and remains to be filled.

A. Future work

- **Field campaigns** A–D: Tables XI–XII.
- **Road-network scale:** corridor–intersection mixed layouts, city-scale multi-LiDAR graphs (journal extension).
- **Unified target framework:** a future T-ITS line may treat sphere/board targets as a *calibration method* for depot-style setups—orthogonal to this paper, where spheres are evaluation rulers only.
- **Online FIM / adaptive planning:** estimator reuse across missions must address session-state teardown before on-line observability auditing.

ACKNOWLEDGMENT

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[TODO: field campaign B: real u_B -bias scatter vs
simulation Fig. 2a]

Fig. 4. Field aspect-bias diagnostic (placeholder).